Backfills should not harm client application queries that are running at the same time. To accomplish that, your backfill operations need to run efficiently and consume few resources. Ideally your server instance is over-provisioned as well, meaning it has a lot of spare compute and disk IO capacity.

From a timing perspective, running backfill queries during a low activity period is a good idea. From an operations and team coordination perspective, capture the SQL being run in a SQL file or script, share it on your team, and conduct the work in a pre-production location first to validate the approach.

An overview of some backfilling concepts is presented below.

- Consider using a *Double Writing* or "Dual Writes" strategy
- Consider a purpose-built intermediate table to assist in the backfill
- Use batched operations to limit resource consumption
- Add a throttle mechanism to slow down row copying if needed. This allows time for Index Maintenance and Replication.
- For a purpose-built intermediate table, create specialized indexes based on how the table is queried. Make the table Unlogged and disable Auto-vacuum

In the next sections you'll explore some of these techniques in greater detail.

Backfilling And Double Writing

Modern engineering organizations are able to perform data migrations online in most cases, to avoid taking down the database server which halts business operations.

This does make the process more challenging and risky compared with a backfill process where the database is fully stopped. This means that online migrations will typically require careful planning and execution.

After introducing a new database column, you may want to plan out multiple phases where first the write operations are set up, and then the reads.

This technique can be called *Double Writing*. Double writing can help as a transitional technique to minimize the time needed in a cutover where reads are switched from the original location to a target location, by ensuring the data population is conducted for both locations at the same time.

Separating Reads and Writes for Backfills

Backfills are straightforward when they can SELECT from one table and UPDATE another table.